

System Proposal

Conference Room Booking and Event Planning System

Prepared For:

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Executive Summary

The Cowboy Hotel needs a new conference booking and event planning system because its current process is manual, paper-based, and difficult to manage. Through the sponsor interview with Jamie Bradson, our team learned that conference room reservations are currently tracked through paper files, sticky notes, and a wall calendar. This has caused problems such as lost information, double-bookings, incorrect room assignments, and poor coordination of catering. Since the hotel has expanded its conference room services, the current process makes it harder for the hotel to manage bookings and fully leverage the expansion. More details about the business need and project value can be found in the System Request section of this deliverable.

Based on the information gathered from the sponsor interview, the proposed system needs to support the main tasks hotel staff perform during the booking and event-planning processes. The system will be used internally by employees, not customers, because hotel staff is responsible for helping customers plan events and manage the details of each reservation. The main functional areas identified include customer management, reservation management, conference room management, catering management, employee management, payment tracking, and report generation. These areas were uncovered through the interview process and helped define what the system needs to do. Further details about these functional areas can be found in the Requirements Definition Document.

The system also needs to follow important business rules identified during the interview. One major requirement is that the system must prevent double-bookings by showing only rooms available for the requested date and time. If two employees try to book the same room at the same time, the first to complete the booking should receive the room, while the second user should receive an error message. The system must also track reservation statuses as unconfirmed, confirmed, or canceled. Canceled reservations should not be deleted, as the hotel wants to keep them for reporting purposes. Since the hotel also works with caterers, managers must acknowledge catering changes or cancellations and communicate them to the caterer. More details about how users complete these tasks can be found in the Use Cases section.

Our team also identified the main data the system needs to store and connect to. Customer records should include customer contact information and, when applicable, organization information. Conference room records should include the room name, size, capacity, and rental fee. Reservation records need to connect customers, rooms, dates, times, prices, statuses, caterers, employees, and payments. The system will not process payments, but it will track payment dates, payment amounts, and the reservation each payment belongs to. This will allow employees to calculate how much a customer still owes. More details about how information moves through the system and how the data is organized can be found in the Process Model and Data Model sections.

The proposed system is also feasible for the Cowboy Hotel. The cost-benefit analysis shows that the system is financially feasible over the three-year analysis period. The project is expected to generate \$662,500 in total benefits and \$464,028 in total costs, resulting in a net benefit of \$198,472 over three years. The project also has a projected ROI of 42.8% and is expected to break even in about 1.3 years. Technically, the system is feasible because it will be web-based and compatible with the hotel's current office PCs and laptops. Operationally, the system is feasible because it will be designed for hotel staff,

support multiple users simultaneously, and help the new booking specialist and the assistant booking specialist manage reservations more efficiently. More details on the project schedule, costs, and feasibility can be found in the Project Work Plan and Feasibility Analysis sections.

The system's recommended acquisition strategy for the Cowboy Hotel is custom development. This would best support the hotel's conference booking and operations, given the system's specifications for reservation management, customer records, catering, payment tracking, and report generation. Specific acknowledgments of catering changes and payment tracking also make custom development the best strategy for the Cowboy Hotel, giving it flexibility and customization.

Overall, the proposed conference room booking and event planning system directly addresses the Cowboy Hotel's current booking problem and supports the hotel's long-term business goals. The system will help reduce manual errors, prevent double-bookings, organize customer and reservation information, improve payment tracking, support catering coordination, and provide useful accounting and marketing reports. Based on the requirements gathered and the analysis completed, the project should move forward to the design phase.

System Request – Conference Room Booking and Event Planning Project

Project Sponsor: Jaime Bradson, Hotel Manager, Cowboy Hotel

Business Need: This project has been initiated to provide Cowboy Hotel with a robust conference room booking and event planning system capable of managing bookings and events for its recent conference room expansion. The current manual booking system is inefficient and error-prone, leading to disorganization and double-booking. A new conference booking system is urgently needed to increase efficiency, better manage data, and begin profiting from its recent expansion.

Business Requirements: The new system will enable staff to easily book conference rooms for guests through a simple, user-friendly interface. It will also provide accurate information, avoid double-booking, and include features to enhance reporting and data entry, creating an easy, efficient, and reliable booking process. Lastly, the system will operate on multiple computers, and the manager will be able to access the booking system from his office.

Business Value: With an organized and efficient process, the new system will effectively support increased conference room bookings. It will also enable Cowboy Hotel to begin advertising its newly added conference rooms and increased space, thereby generating more bookings. Additionally, the hotel will realize increased hotel room revenue from additional reservations by conference and event attendees.

The estimated benefits are:

- **\$78,125** increase in conference room booking sales during year 1
- **\$178,125** increase in conference room booking sales by the end of year 3
- **\$50,000** increase in hotel room revenue from conference guests staying at the hotel during conferences and events

Special Issues or Constraints:

- The project addresses an urgent business need. The project is critical for effectively managing and profiting from the recent expansion of Cowboy Hotel's conference rooms. Therefore, it must be completed promptly.
- The project must adhere to the **\$60,000** budget set by the hotel while meeting all business requirements.

Project Work Plan

Task ID	Task Name	Assigned To	Duration (days)	Est. Start Date	Est. Finish Date	Actual Start Date	Actual Finish Date	Duration Variance	Dependency	Status
<u>1. Planning Phase</u>										
1.1	Develop a project work plan	Aarav	4	17-Jan	20-Jan	29-Jan	1-Feb	0		Completed
1.2	Define project scope and objectives	Team	4	20-Jan	23-Jan	5-Feb	8-Feb	0	1.1	Completed
1.3	Identify stakeholders and sponsor	Team	3	23-Jan	25-Jan	10-Feb	12-Feb	0	1.2	Completed
<u>2. Analysis Phase</u>										
2.1	Develop system request	Rose	3	25-Jan	27-Jan	18-Mar	20-Mar	0	1.3	Completed
2.2	Create cost-benefit analysis	Celine	9	27-Jan	4-Feb	10-Mar	17-Mar	-1	2.1	Completed
2.3	Gather system requirements	Team	9	7-Feb	15-Feb	10-Mar	20-Mar	+2	2.2	Completed
2.4	Create use cases	Team	9	21-Feb	1-Mar	12-Mar	22-Mar	+2	2.3	Completed
2.5	Perform process & data modeling	Team	9	28-Feb	8-Mar	22-Feb	10-Mar	+8	2.4	Completed
<u>3. Design Phase</u>										
3.1	Design system architecture	Team	9	21-Mar	29-Mar	4-Apr	10-Apr	-2	2.4	Completed
3.2	Design user interface mockups	Team	9	4-Apr	12-Apr	10-Apr	18-Apr	0	3.1	Completed
3.3	Design database & program structure	Team	9	11-Apr	19-Apr	18-Apr	25-Apr	-1	3.2	Completed
<u>4. Implementation Phase</u>										
4.1	Develop & test booking system	Team	22	15-May	5-Jun				3.3	Open
4.2	Deploy system & train users	Team	10	6-Jun	15-Jun				4.1	Open

Feasibility Analysis

Cost-Benefit Analysis					
	2026	2027	2028	2029	Total
Benefits					
Increased conference room sales	78,125	162,500	178,125	93,750	512,500
Increased hotel room sales	25,000	50,000	50,000	25,000	150,000
Total Benefits	103,125	212,500	228,125	118,750	662,500
Development Costs					
Server licenses	10,000	0	0	0	10,000
Development labor	40,000	0	0	0	40,000
2 computers @ \$1,724 each	3,448	0	0	0	3,488
Server Software	880	0	0	0	880
2 Monitors @ \$165 each	330	0	0	0	330
2 Keyboard @ \$25 each	50	0	0	0	50
2 Mice @ \$20 each	40	0	0	0	40
Tower Server	1900	0	0	0	1900
Total Development Costs	56,648	0	0	0	56,648
Operational Costs					
Software Maintenance	200	200	200	280	880
User training	500	1,000	1,000	500	3,000
Booking Specialist	32,500	65,000	65,000	32,500	195,000
Assistant Booking Specialist	30,000	60,000	60,000	30,000	180,000
Misc.	4,750	9,500	9,500	4,750	28,500
Total Operational Costs	67,950	135,700	135,700	68,030	407,380
Total Costs	124,598	135,700	135,700	68,030	464,028
Total Benefits – Total Costs	(19,749)	76,800	92,425	50,720	198,472
Cumulative Net Cash Flow	(19,749)	57,051	149,476	198,472	
Return on Investment (ROI)	42.8%				
Break-Even Point	1.3 years				

Economic Feasibility

Based on the cost-benefit analysis, the proposed system is financially feasible for the Cowboy Hotel. Since the system is expected to go live around July 2026, our team prorated the 2026 benefits and operational costs for half of the year. The 2029 benefits and operational costs were also prorated, so the analysis still covers a full three-year operating period. Over the three-year period, the system is expected to generate \$662,500 in total benefits and \$464,028 in total costs, resulting in a net benefit of \$198,472. The project also has a projected ROI of 42.8% and is expected to break even in approximately 1.3 years. This shows that the system would provide enough financial benefit to support moving forward with the project.

Technical Feasibility

The Cowboy Hotel is currently operating using Windows 11 and Microsoft Office. The hotel plans to add two more computers to support its operations. The system will use functionalities such as reservation management, customer records, payment tracking, catering management, and booking report generation. These could be developed using databases and a web-based system, so no complicated software needs to be installed. The proposed system will not include advanced technology, as its features are sufficient. With the allocated budget and resources, the proposed system will be technically feasible.

Operational Feasibility

The Cowboy Hotel currently handles reservations and event planning manually, leading to errors such as double-booking and inefficient operations. The proposed system will automate key tasks, like reservation creation, customer management, payment tracking, reporting, and catering management. This would mean staff could focus on other parts of the Cowboy hotel's operations, while the system handles operations. The system will have a user-friendly interface, making it easily accessible to staff and easier to understand. Hiring a Booking Specialist and an Assistant Booking Specialist will ensure competent staff manage the system. This proposed system will assure the organization's workflow is efficient, improving service and minimizing errors.

Requirements Definition Document

FUNCTIONAL REQUIREMENTS

1. Customer Management

- 1.1 The system allows employees to create customer records.
- 1.2 The system will store customer information (name, organization name, address, email, phone number).
- 1.3 The system allows employees to view customer reservation records.
- 1.4 The system allows employees to update customers' information.
- 1.5 The system will prevent duplicate customer records from being created.

2. Reservation

- 2.1 The system allows employees to create reservations for customers.
- 2.2 The system will store reservation details (customer, start date/time, end date/time, total price, room quantity, status).
- 2.3 The system will enable employees to assign one or more available conference rooms to a reservation.
- 2.4 The system will store and display the specific conference rooms assigned to each reservation.
- 2.5 The system will track reservation status (UNCONFIRMED, CONFIRMED, CANCELLED).
- 2.6 The system will set a reservation to UNCONFIRMED when first created.
- 2.7 The system will update the reservation status to CONFIRMED when a down payment is recorded.
- 2.8 The system will update the reservation status to CANCELLED when an employee cancels a reservation.
- 2.9 The system will allow employees to modify, update, or cancel reservations.
- 2.10 The system will enable employees to make changes to catering or cancellations of caterers for specific reservations.
- 2.11 The system will require manager acknowledgment before saving any catering modification or cancellation.
- 2.12 The system will maintain a record of which employee created or last modified each reservation, along with a timestamp.
- 2.13 The system will keep all records of canceled reservations.
- 2.14 The system will allow employees to view the full details of a reservation.

3. Conference Room

- 3.1 The system will store all conference room information (room name, room size, capacity, rental fee)
- 3.2 The system will show the available conference rooms on the date and time customers need.
- 3.3 The system will check room availability before authorizing a reservation.
- 3.4 The system will notify the user if a selected room becomes unavailable because it has been reserved.

4. Catering

- 4.1 The system will allow employees to create caterer records.
- 4.2 The system will store all caterer information (name, email address, phone number).
- 4.3 The system will allow employees to modify a caterer record.

- 4.4 The system will store updated caterer records.
- 4.5 The system will enable caterers to be assigned to reservations.
- 4.6 The system will store the assigned caterer for each reservation.

5. Payment

- 5.1 The system will enable employees to record a down payment and additional payments for a reservation.
- 5.2 The system will store payment records (payment date, payment amount, reservation)
- 5.3 The system enables employees to view payment histories for each reservation.

6. Reporting

- 6.1 The system will report revenue from each room.
- 6.2 The system will report revenue within a selected time period.
- 6.3 The system will show marketing reports of the most frequently reserved conference rooms.
- 6.4 The system will report the rarely used conference rooms.

7. Employees

- 7.1 The system will store employee information (name, email, phone number, role).
- 7.2 The system will require employees to log in with a username and password before accessing the system.
- 7.3 The system will support employee and manager roles, with managers having the additional permission to acknowledge catering modifications and cancellations.

NONFUNCTIONAL REQUIREMENTS

1. Operational

- 1.1 The system will be accessible to staff within the hotel.
- 1.2 The system will be compatible with Microsoft Windows 11 and run on the PCs and laptops using Google Chrome and Microsoft Edge browsers used by hotel staff.
- 1.3 The system will provide a simple and user-friendly interface to support efficient data entry by hotel staff.

2. Performance

- 2.1 The system will support multiple users simultaneously.
- 2.2 The system will reflect room availability changes within three seconds after a reservation is created or canceled.
- 2.3 The system will process reservation updates and cancellations within three seconds.
- 2.4 The system will remain operational and responsive during periods of peak usage by staff.

3. Security

- 3.1 The system will require employee login authentication before granting access.
- 3.2 The system will securely store customer personal information and payment records.
- 3.3 The system will maintain records of employee activity on reservation records for security auditing purposes.

4. Political/Cultural

- 4.1 The system must comply with the hotel's internal policies for reservations and payment records, including policies for handling double-booking.
- 4.2 The system must protect all customer and payment data in compliance with applicable data privacy regulations and hotel data protection standards.

- 4.3 The system must retain all historical data indefinitely unless otherwise directed by hotel management.
- 4.4 The system must enforce manager acknowledgment for all catering modifications and cancellations in accordance with hotel policy.
- 4.5 The system will display and store all monetary values in U.S. dollars only.
- 4.6 The system will use standard U.S. currency formatting and U.S. date formats (MM/DD/YYYY) throughout the application.

Use Cases

Use Case Name: Search conference room availability	ID: UC-1	Priority: High
Actor: Booking Specialist		
Description: The booking specialist enters the requested date(s) to search for available conference rooms and reviews the room details to assist a customer in selecting a room for reservation		
Trigger: A customer requests conference room availability		
Type: External		
Preconditions: 1. The conference booking system is online 2. The booking specialist is authenticated by logging in to their user account		
Normal Course: 1.0 Find available conference room(s) 1. The booking specialist enters the requested date(s) 2. The system displays available conference rooms and their available time slots for the selected date(s) 3. The specialist chooses a conference room from the list to view 4. The system displays detailed room information for the chosen room 5. The booking staff member reviews the room information with the customer 6. The customer decides whether to proceed with a reservation		
Postconditions: 1. Detailed information for a selected room is displayed 2. The selected room is available for reservation if the customer decides to proceed		
Exceptions: - E1. No conference rooms are available for the selected date(s) and time(s) (occurs at Step 2) 1. The system displays a message indicating that no rooms are available 2. The booking specialist revises the search criteria or exits		

Use Case Name: Create a reservation	ID: UC-2	Priority: High
Actor: Booking Specialist		
Description: This use case describes how a booking specialist creates a new reservation for one or more conference rooms on behalf of a customer. The system allows the booking specialist to enter customer information, check room availability, assign rooms, associate catering if needed, and calculate the total cost. The reservation is initially unconfirmed and becomes confirmed once a down payment is recorded.		
Trigger: A customer contacts the facility (by phone or in person) requesting to reserve one or more conference rooms for an event		
Type: External		
Preconditions: 1. The booking specialist is logged into the system. 2. Room, pricing, and scheduling data already exist in the system. 3. The system is operational.		
Normal Course: 1.0 Create a reservation 1. The booking specialist begins a new reservation. 2. The system prompts for customer and event information. 3. The booking specialist enters the required customer details. 4. The booking specialist searches for room availability for the requested date and time. 5. The system displays only available rooms. 6. The booking specialist selects one or more rooms. 7. The booking specialist enters pricing details and applies any applicable discounts. 8. The booking specialist indicates whether catering is required. 9. The system calculates the total reservation cost. 10. The booking specialist saves the reservation. 11. The system stores the reservation with a status of "Unconfirmed".		
Postconditions: 1. A new reservation is saved in the system. 2. The reservation status is set to "Unconfirmed". 3. Assigned rooms, date/time, pricing, and customer details are stored. 4. The reservation is available for future payment tracking and reporting.		
Exceptions: E1: Room no longer available (occurs at Step 4) • If another user reserves the room before saving: 1. The system displays an error message indicating the room is no longer available. 2. The booking specialist selects a different room or restarts the search.		

Use Case Name: Modify customer reservation	ID: UC-3	Priority: High
Actor: Booking Specialist		
Description: This use case describes how a booking specialist modifies an existing reservation.		
Trigger: A customer requests a change to a reservation.		
Type: External		
Preconditions: 1. Booking Specialist is authenticated. 2. The booking system is available online. 3. The modification request is for an existing reservation. 4. The new room date is available. 5. The requested room is available on the new day		
Normal Course: 1.0 Modify reservation 1. The booking specialist locates an existing reservation. 2. The booking specialist inputs the requested changes to the reservation. 3. The system verifies that the changes don't conflict with another reservation. 4. The system updates the reservation with any price changes. 5. The system prompts for manager acknowledgment for catering changes. 6. The manager acknowledges catering changes. 7. The system stores the updated reservation.		
Postconditions: 1. Updated reservation is saved and stored. 2. Availability is updated. 3. Amount due accurately reflects updated reservation.		
Exceptions: - E1. The modifications are not valid (occurs at step 3) 1. If the requested modifications are not valid, the system will notify the booking specialist, and new options may be entered.		

Use Case Name: Cancel a customer reservation	ID: UC-4	Priority: High
Actor: Booking Specialist		
Description: This use case describes how a booking specialist cancels an existing reservation.		
Trigger: A customer requests a cancellation of a reservation.		
Type: External		
Preconditions: 1. Booking specialist is authenticated. 2. Booking system is available online. 3. The cancellation request is for an existing reservation.		
Normal Course: 1.0 Cancel reservation 1. Booking specialist locates the existing reservation. 2. Booking specialist begins cancellation. 3. The system displays cancellation details and the refund policy. 4. The system prompts for manager acknowledgment for catering cancellation. 5. Manager acknowledges catering cancellation. 6. System marks the reservation as canceled. 7. System updates room availability.		
Postconditions: 1. The reservation is marked as canceled in the system. 2. Availability is updated. 3. The outstanding balance is refunded to the customer.		
Exceptions: E1. Reservation cannot be found 1. The system displays a message that the reservation was not found 2. The booking specialist verifies the reservation information with the customer 3. The booking specialist revises the search criteria or exits the use case		

Use Case Name: Create a caterer	ID: UC-5	Priority: Medium
Actor: Booking Specialist		
Description: This use case describes how a booking specialist creates and stores a new caterer in the system. The system allows the booking specialist to enter the caterer's details, such as name and contact information, so the caterer can be assigned to reservations later.		
Trigger: A booking specialist needs to add a new caterer to the system.		
Type: External		
Preconditions: 1. The booking specialist is logged into the system. 2. The system is operational.		
Normal Course: 1.0 Create a caterer 1. The booking specialist selects the option to create a new caterer. 2. The system prompts for caterer information. 3. The booking specialist enters the required caterer details (name and contact information). 4. The system validates the entered information. 5. The booking specialist saves the caterer. 6. The system stores the caterer.		
Postconditions: 1. A new caterer is saved in the system. 2. The caterer is available to be assigned to reservations.		
Exceptions: E1. Caterer record cannot be saved because of a system or database error 1. The system displays an error message indicating the caterer could not be saved 2. The booking specialist exits the use case or tries again later		

Use Case Name: Modify a caterer	ID: UC-6	Priority: Medium
Actor: Booking Specialist		
Description: This use case describes how a Booking Specialist modifies an existing caterer's information in the system so that caterer records remain accurate and up to date.		
Trigger: A Booking Specialist needs to update the information of an existing caterer in the system.		
Type: External		
Preconditions: 1. The booking specialist is logged in to the system. 2. The system is operational. 3. The caterer record to be modified is already in the system.		
Normal Course: 1.0 Modify a caterer 1. The booking specialist selects the option to modify an existing caterer. 2. The system prompts the booking specialist to search for the caterer. 3. The booking specialist enters search criteria to locate the caterer. 4. The system retrieves and displays the matching caterer record. 5. The booking specialist updates the necessary caterer details (name, contact information). 6. The booking specialist saves all the changes. 7. The system validates the updated information. 8. The system stores the updated caterer record.		
Postconditions: 1. The caterer's record reflects the updated information. 2. The updated caterer is available to be assigned to reservations.		
Exceptions: E1. Caterer not found (occurs at Step 4) • If no caterer record matches the search criteria: 1. The system displays a message indicating no matching caterer was found. 2. The booking specialist refines the search criteria and tries again, or ends the use case. E2. Missing or invalid information (occurs at Step 7) • If required fields are missing or invalid: 1. The system displays a message indicating which fields need correction. 2. The booking specialist corrects the information and resubmits.		

Use Case Name: Add a new customer record	ID: UC-7	Priority: High
Actor: Booking Specialist		
Description: This use case describes how a booking specialist adds a new customer's information into the system so that the specialist can create conference room reservations for the customer.		
Trigger: A customer requests a conference room booking, and their information does not already exist in the system.		
Type: External		
Preconditions: 1. The booking specialist is logged in to the system 2. The application and its components are operational and available		
Normal Course: 1.0 Add a new customer record 1. The booking specialist selects the option to add a new customer. 2. The system prompts for customer information. 3. The booking specialist enters customer information (name, phone number, email, organization, and address) 4. The booking specialist saves the customer's information. 5. The system validates customer information. 6. The system saves the new customer record. 7. The system displays a confirmation screen indicating the customer record has been successfully created.		
Postconditions: 1. A new customer record is created and saved within the system. 2. Customer information is available for future bookings.		
Exceptions: - E1. Customer already exists in the application (occurs at step 5) 1. The application displays a message stating that the customer already exists. 2. The booking specialist exits, and the use case ends.		

Use Case Name: Modify a customer record	ID: UC-8	Priority: High
Actor: Booking Specialist		
Description: This use case describes how a booking specialist updates an existing customer's information in the system to ensure customer records remain accurate and up to date for reservation and billing purposes		
Trigger: A customer requests a change to their personal or organizational information, or the booking specialist identifies incorrect or outdated customer data		
Type: External		
Preconditions: 1. Booking specialist is authenticated and logged into the system 2. Customer record already exists in the system 3. System is operational		
Normal Course: 1. Booking specialist searches for an existing customer record 2. System displays matching customer records 3. Booking specialist selects the appropriate customer profile 4. System displays the customer's current information 5. Booking specialist updates the required customer fields 6. System validates the updated information 7. Booking specialist saves the updated customer record 8. System stores the updated customer information and confirms the changes made		
Postconditions: 1. Customer record is updated and stored in the system 2. Updated customer information is available for future reservations and reporting		
Exceptions: E1. Customer record not found (occurs at step 2) 1. The system displays a message indicating no matching customer was found 2. The booking specialist refines the search criteria or ends the use case E2. Invalid customer information (occurs at step 6) 1. The system displays an error message indicating missing fields 2. Booking specialist corrects the information		

Use Case Name: Track customer payments	ID: UC-9	Priority: High
Actor: Booking Specialist		
Description: This describes how a booking specialist would record a customer's payment for an existing conference room. When payment is recorded, the system updates the amount paid and calculates the remaining balance. If this is the first payment received, the reservation status changes from UNCONFIRMED to CONFIRMED .		
Trigger: Booking specialist receives payment from a customer and records it in the system.		
Type: External		
Preconditions: 1. A reservation exists in the system. 2. The total reservation amount has already been recorded. 3. The booking specialist has access to the reservation information.		
Normal Course: 1.0 Track customer payments 1. The booking specialist searches for the reservation. 2. System retrieves and displays the reservation details (reservation status, total amount, amount paid). 3. The booking specialist enters the payment amount and the payment date. 4. System records the payment associated with the reservation. 5. System updates the amount paid for the reservation. 6. The system calculates the remaining balance based on total charges and payments received. 7. If no previous payments exist, the system changes the reservation status from UNCONFIRMED to CONFIRMED.		
Alternative Course: Reservation already has prior payment (Step 7) 1. The system determines that a previous payment has already been recorded. 2. The system records the payment. 3. Reservation status remains unchanged. 4. Use case ends.		
Postconditions: 1. The payment is recorded and saved with the associated reservation. 2. The total amount paid is updated. 3. The reservation status is updated to "Confirmed" if this is the first payment.		
Exceptions: E1. Payment Exceeds Remaining Balance (occurs at step 3) 1. The system displays error message indicating that the payment exceeds the remaining balance. 2. The system prevents the payment from being recorded. 3. Use case ends.		

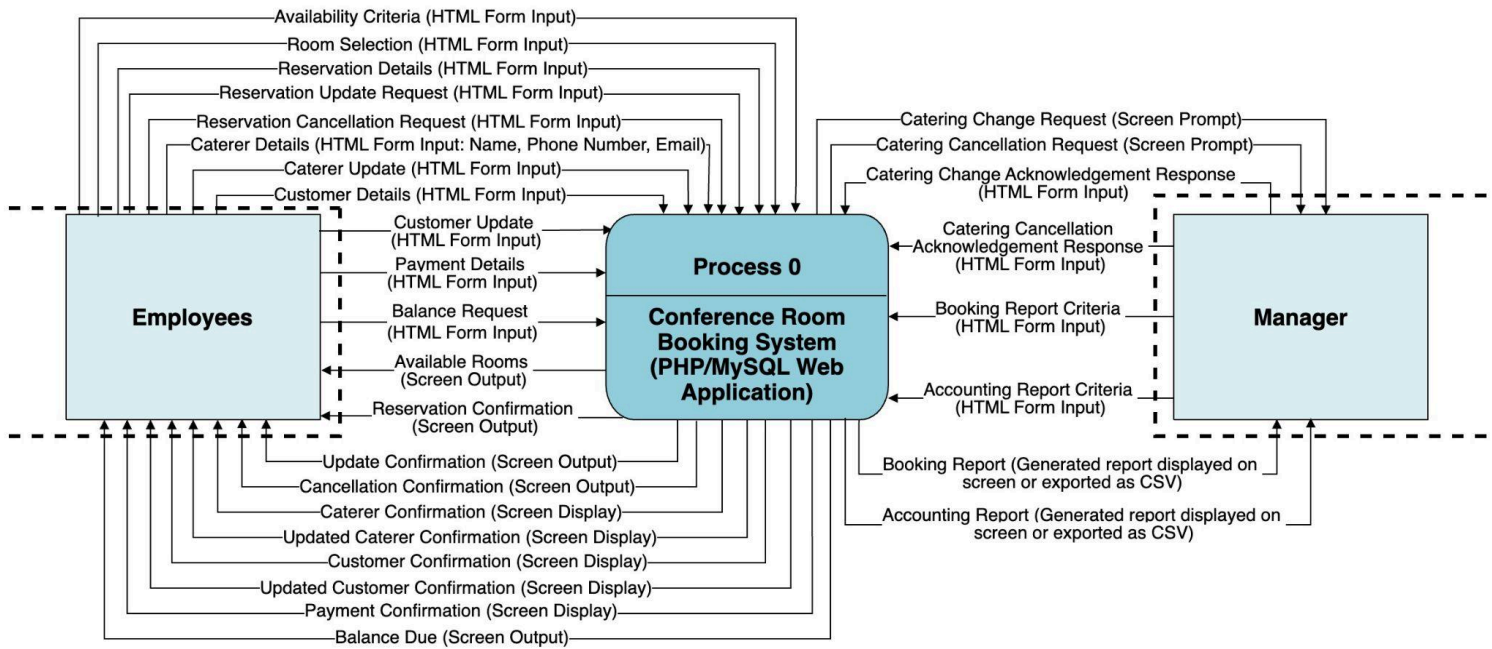
Use Case Name: Calculate a customer's balance due	ID: UC-10	Priority: High
Actor: Booking Specialist		
Description: This use case describes how the system calculates the remaining balance for a customer's reservation by subtracting all recorded payments from the total reservation cost. This allows the staff to respond to customer inquiries regarding outstanding balances due.		
Trigger: A booking specialist or customer requests the remaining balance for an existing reservation		
Type: External		
Preconditions: 1. Booking specialist is authenticated and logged into the system 2. Reservation exists in the system 3. Reservation has a total cost recorded 4. Payment records exist for the reservation		
Normal Course: 1. Booking specialist searches for a reservation using customer or reservation ID 2. System retrieves and displays reservation details 3. System retrieves all payment records associated with reservation 4. System calculates the total amount paid 5. System subtracts total payments from total reservation cost 6. System displays the remaining balance due 7. Booking specialist reviews the balance with customer		
Postconditions: 1. Customer's remaining balance is calculated and displayed 2. No data is modified in the system		
Exceptions: E1. Reservation not found (occurs at step 1) 1. The system displays an error message indicating that no matching reservations exist 2. Booking specialist corrects the search or exits the use case		

Use Case Name: Generate a booking report	ID: UC-11	Priority: Medium
Actor: Hotel Manager		
Description: This use case describes how the hotel manager generates a booking report using the Cowboy Hotel conference management system to review reservation activity, room usage, and booking trends.		
Trigger: Hotel Manager requests a booking report		
Type: External		
Preconditions: 1. Hotel Manager is logged into the system 2. Reservation records exist in the system 3. System is operational		
Normal Course: 1.0 Generate a booking report 1. Hotel Manager begins generating a booking report. 2. System displays report options and filters. 3. Hotel Manager enters report criteria such as date range, room type, or event category. 4. System validates the entered criteria. 5. Hotel Manager submits the report request. 6. System retrieves reservation data from the reservation records. 7. System compiles the booking report based on the selected criteria. 8. System displays the completed booking report. 9. Hotel Manager reviews the booking report. 10. System provides options to print or export the report.		
Postconditions: 1. Booking report is generated and displayed 2. Hotel Manager may print or export the report		
Exceptions: E1. Invalid report criteria (occurs at step 3) 1. System displays an error message indicating invalid or incomplete criteria. 2. Hotel Manager corrects the report criteria. E2. No booking data found (occurs at step 8) 1. System displays a message that no booking data matches the selected criteria. 2. Hotel Manager may enter different criteria or cancel the request.		

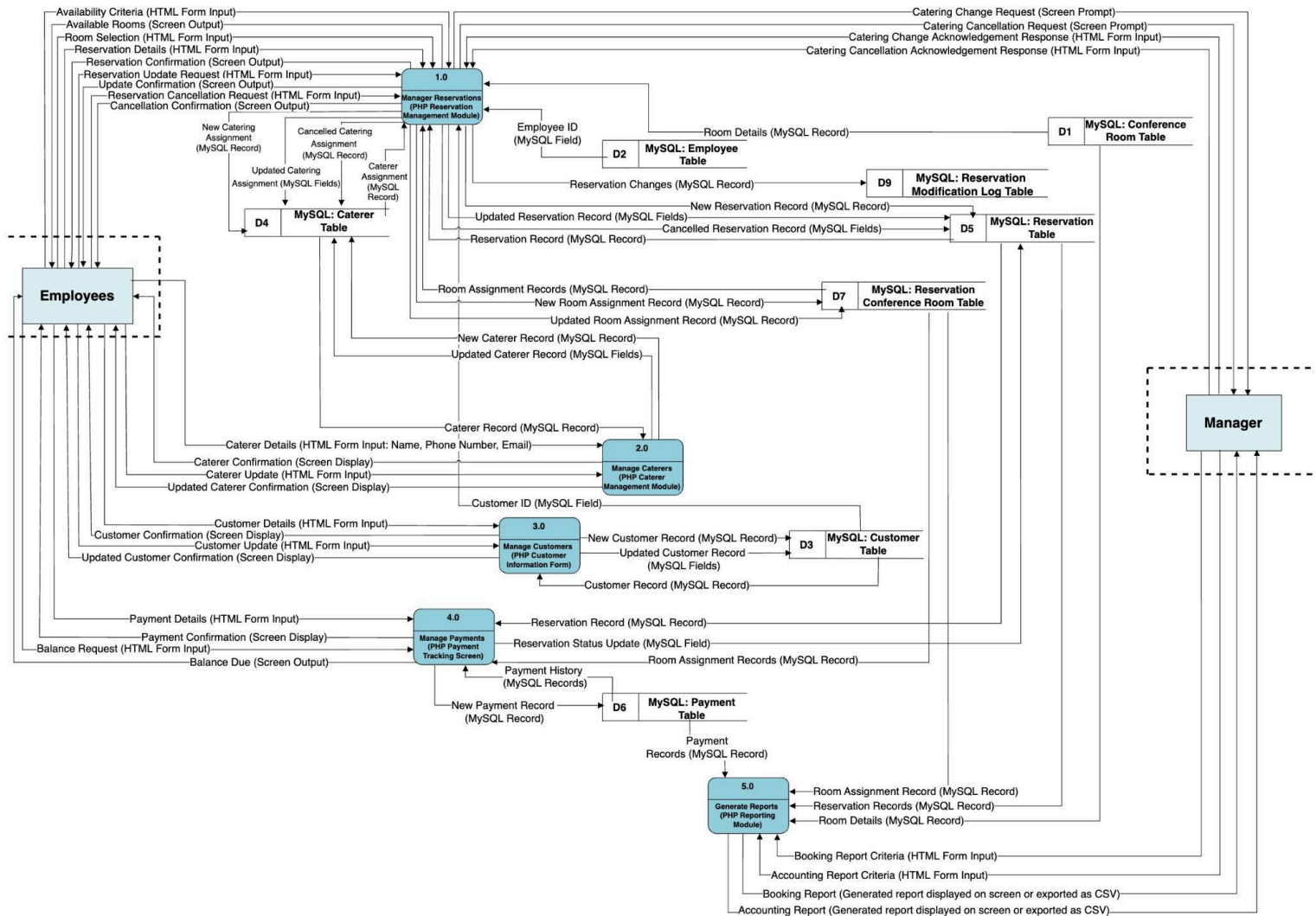
Use Case Name: Generate an accounting report	ID: UC-12	Priority: Medium
Actor: Hotel Manager		
Description: This use case describes how the hotel manager generates an accounting report using the Cowboy Hotel conference management system to review payment activity, deposits, balances, and revenue information.		
Trigger: Hotel Manager requests an accounting report		
Type: External		
Preconditions: 1. Hotel Manager is logged into the system. 2. Payment and reservation records exist in the system. 3. System is operational.		
Normal Course: 1.0 Generate an accounting report 1. Hotel Manager begins generating an accounting report. 2. System displays report options and filters. 3. Hotel Manager enters report criteria such as date range, payment status, or event type. 4. System validates the entered criteria. 5. Hotel Manager submits the report request. 6. System retrieves payment and reservation data from the payment and reservation records. 7. System compiles the accounting report based on the selected criteria. 8. System displays the completed accounting report. 9. Hotel Manager reviews the accounting report. 10. System provides options to print or export the report.		
Postconditions: 1. Accounting report is generated and displayed. 2. Hotel Manager may print or export the report.		
Exceptions: E1. Invalid report criteria (occurs at step 3) 1. System displays an error message indicating invalid or incomplete criteria. 2. Hotel Manager corrects the report criteria. E2. No accounting data found (occurs at step 8) 1. System displays a message that no accounting data matches the selected criteria. 2. Hotel Manager may enter different criteria or cancel the request.		

Physical Process Model

Context Diagram

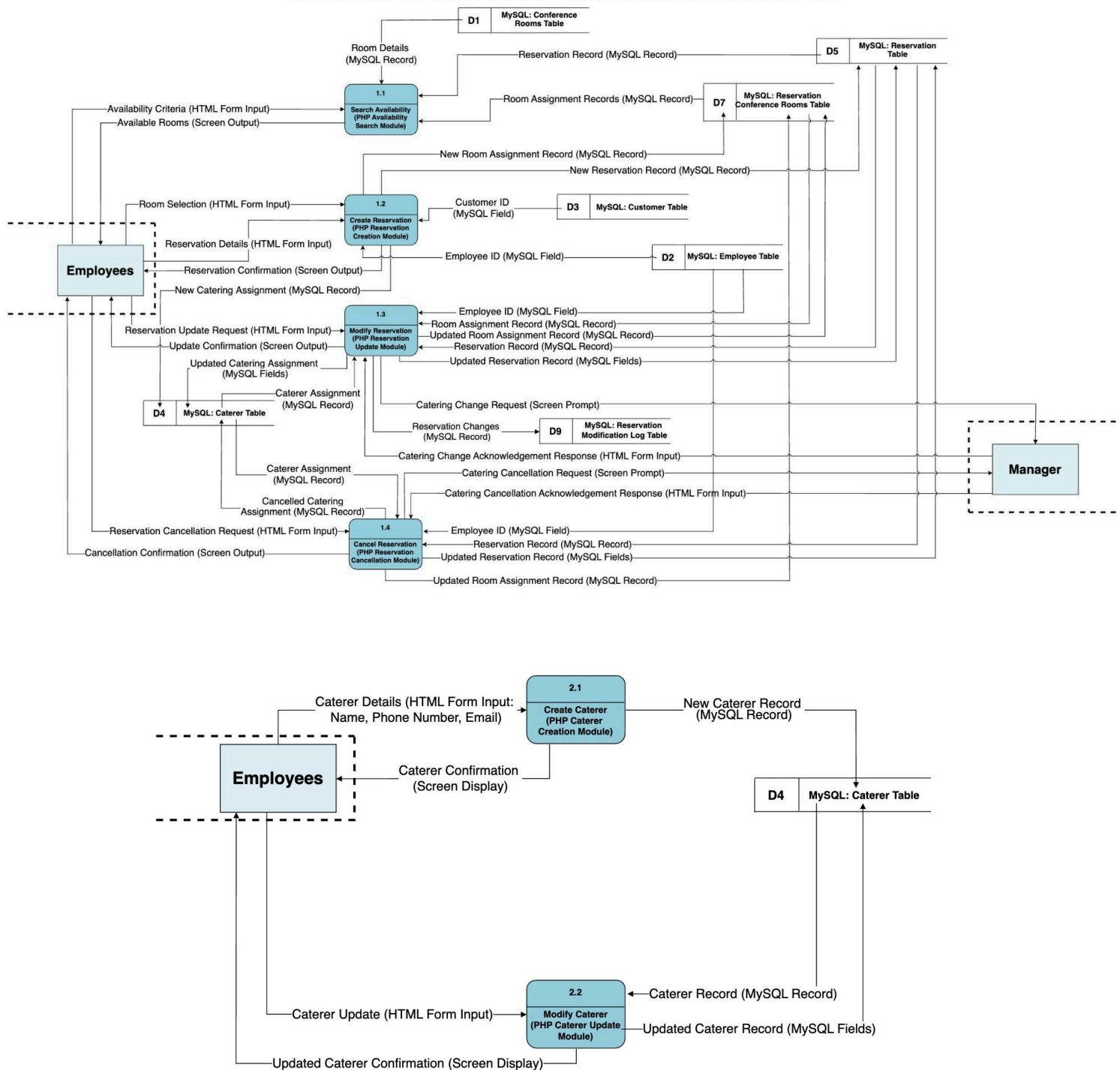


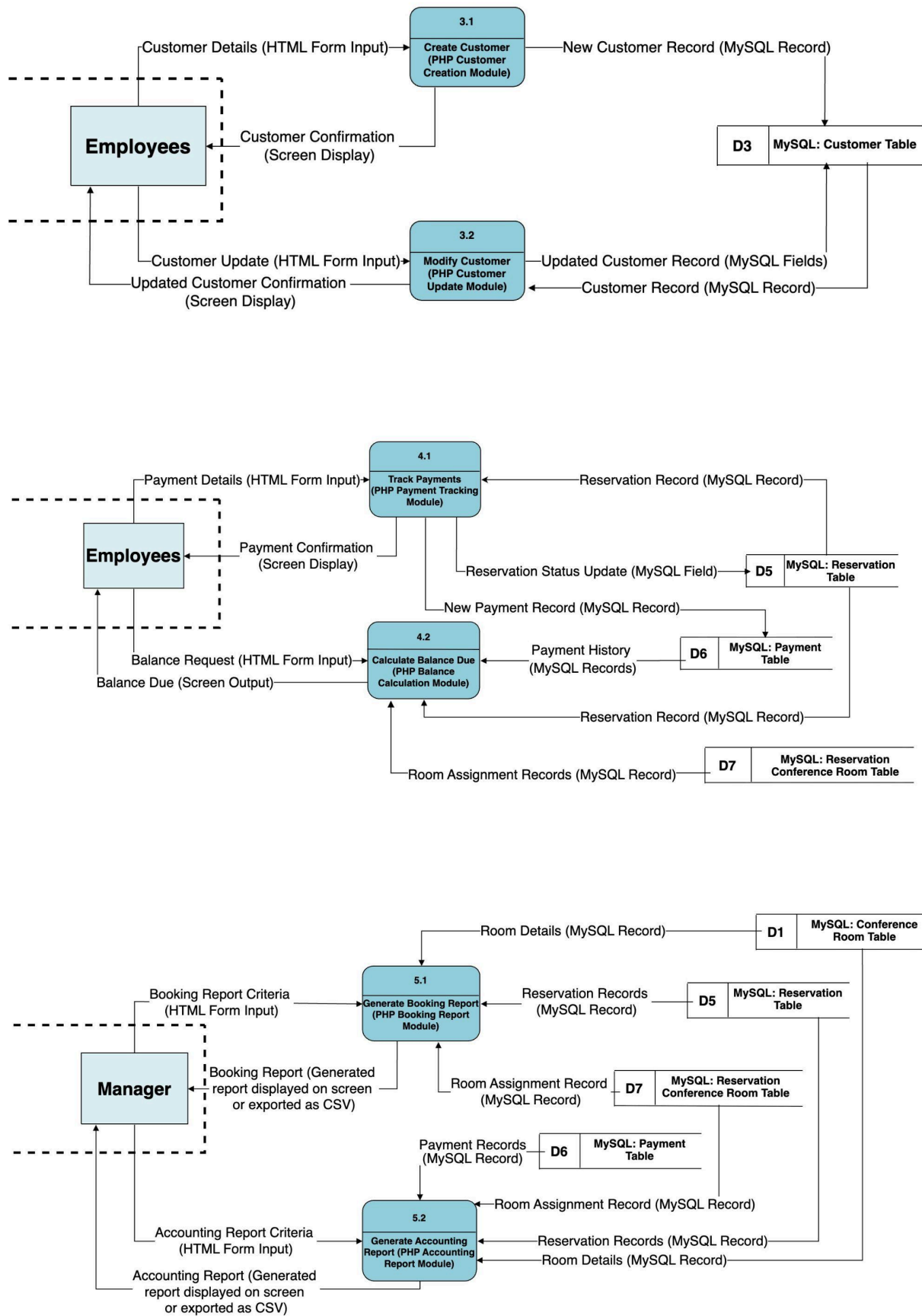
Level 0 Diagram



Level 1 Diagrams

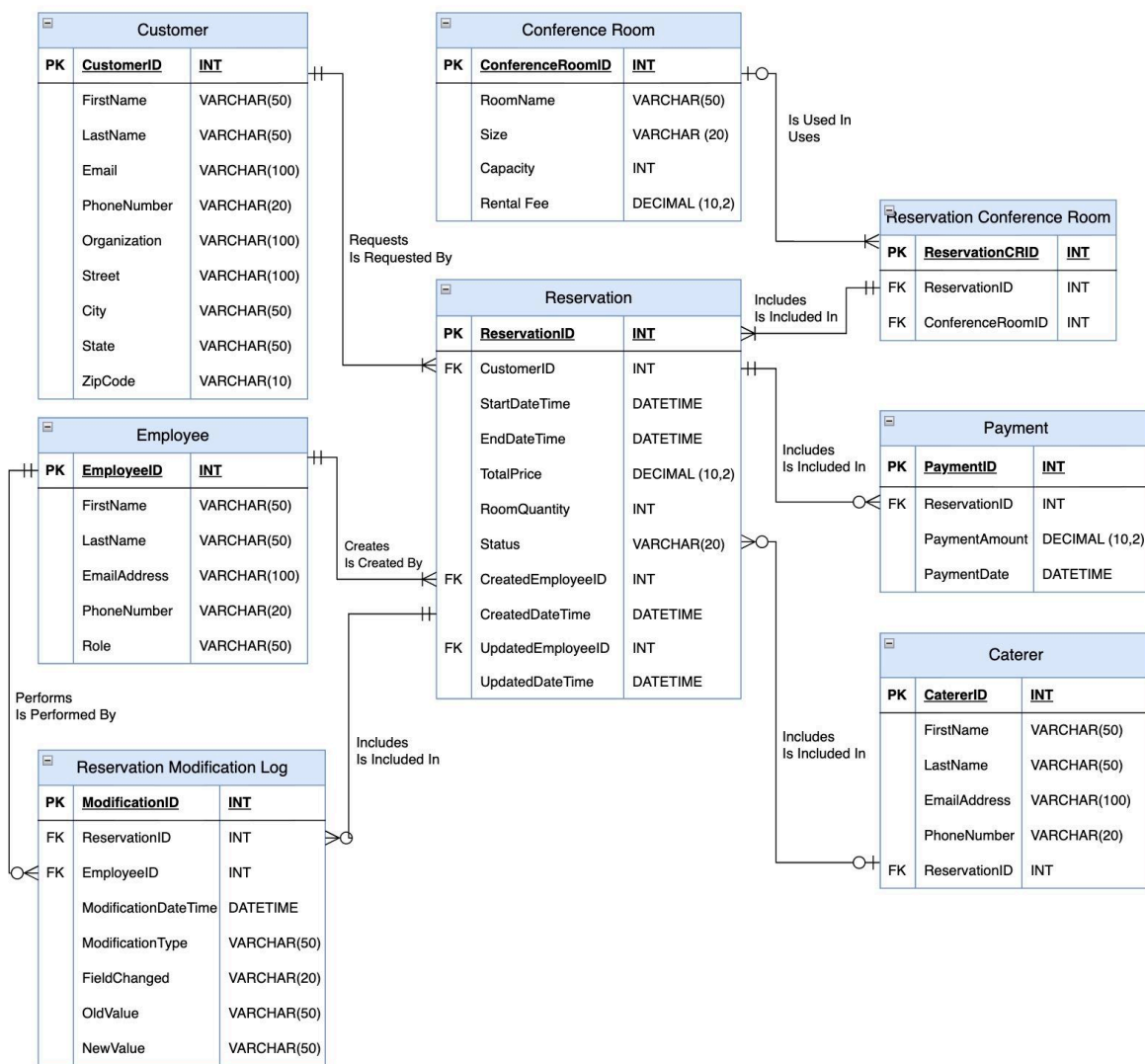
Note: The Caterer table is located at the bottom left to improve readability and reduce excessive line overlap. Additionally, some processes use the same data flows as inputs, such as reservation records; therefore, more data flows will be observed, but these represent detailed implementations of existing higher-level flows rather than entirely new flows.





Physical Data Model

E-R Diagram



HW-SW Specification

	Standard Client	Standard Server
Operating System	• Windows 11 Pro	• Windows Server 2025 Standard
Special Software	• Office Professional 2021 (current equivalent for 2019 version) • Microsoft Edge (or Chrome)	• IIS • MySQL Server 8.0 • ASP.NET Core
Hardware	• Intel Core i5 • 8GB RAM • 256GB SSD Link: Windows Desktop • 22" LED Monitor • Keyboard • Mouse	• Intel Xeon E Processor Series • 16GB RAM • 2TB HDD Link: Tower Server
Network	• 1 Gbps Ethernet	• 1 Gbps Ethernet

Updated Cost/Benefit Analysis

Estimated Cost of Each Piece of Hardware:

- Desktop - \$862
 - Windows 11 Pro
 - Intel Core i5
 - 8GB RAM
 - 256GB SSD
- Monitor - \$165
 - 22" LED Monitor
- Tower Server - \$1,900
 - Windows Server 2025 Standard
 - Intel Xeon E Processor Series
 - 16GB RAM
 - 2TB HDD
- Keyboard - \$25 each
- Mouse - \$20 each

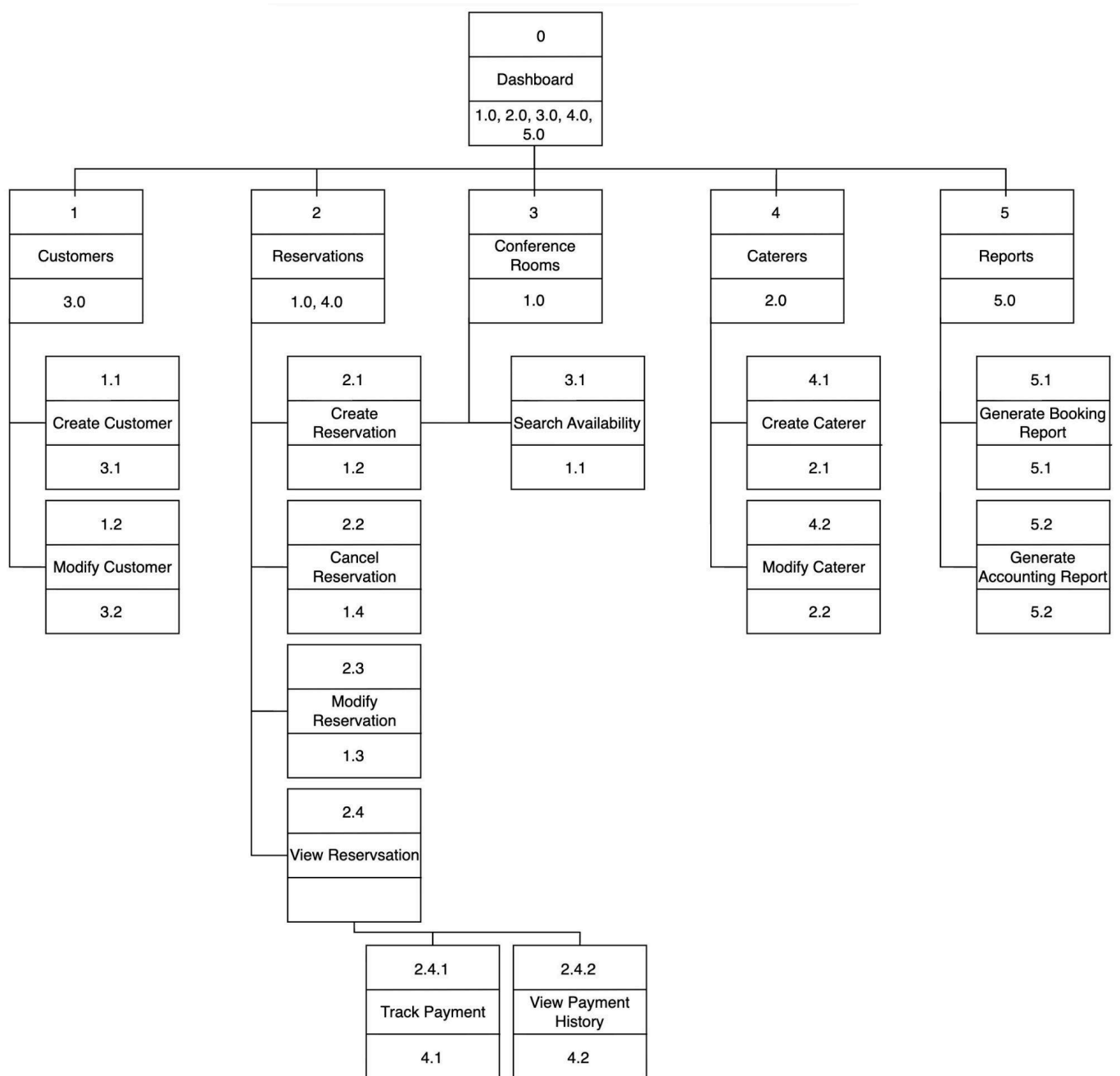
Estimated Total Cost of Hardware:

- Estimated Total Cost of Hardware = **\$4,044**
 - Desktop (2) - \$1724
 - Monitor (2) - \$330
 - Tower Server (1) - \$1,900
 - Keyboard (2) - \$50
 - Mouse (2) - \$40

Estimated Cost of Each Piece of Software:

- Office Professional 2021 - \$440
- Microsoft Edge (or Chrome) - \$0
- IIS - \$0
- ASP.NET Core - \$0
- MySQL Server 8.0 - \$0

Interface Structure Diagram



User Interface

Generate a Report (Wireframe Diagram)

Site Name	Dashboard	Customers	Reservations	Rooms	Caterers	Reports	User Account Icon
Reports / Generate Report							
Generate Report					Export CSV	Print Report	
Filters							
Report Type		Start Date		End Date		Generate	
Revenue By Room 							
		This Month		Last 3 Months		Year-To-Date	
Report Content							
Page Indicator						Page Navigation Bar	

Cancel Reservation (Screen Prototype)

Hotel Booking System

John Smith
Booking Specialist

Cancel Reservation

Search, Review, Cancel

Search Reservation

Reservation ID
#####

Customer Name
XXXXX XXXXX

Phone Number
(###) ### ####

Booking Date
mm/dd/yyyy

Search

Reservation Details

Reservation ID:	#####	Room Type:	Meeting Room 1
Customer Name:	XXXXX XXXXX	Check-in:	6:00 AM
Phone Number:	(###) ### ####	Check-out:	3:00 PM
Booking Date:	mm/dd/yyyy	Status:	Active

Cancellation Details

Reason for Cancellation:
...

Date Cancelled: mm/dd/yyyy
Refund Amount: \$\$\$\$

Appendix A

Interview Report

Person Interviewed: Jaime Bradson (Hotel Manager)

Interviewers: Rose Garcia, Celine Vong, Nathan Ng, Aarav Rawal, Brandon Martin Garcia, Angel Nolasco Vega, Andre Sibal

Purpose of Interview:

- To identify challenges with the current system and their impact on the company's revenue.
- To determine system requirements for the new conference room booking system.
- To uncover any additional specific or unique system requests.

Summary of Interview:

- The current process for booking conference rooms is entirely manual and paper-based, making it difficult to keep track of bookings and leading to errors, such as:
 - Lost and incorrect information.
 - Double-bookings due to poor coordination and tracking.
 - Human error and general careless mistakes.
- The new system should eliminate double-booking issues and display each room's correct availability.
 - Double-booking will be resolved on a “first-come, first-served” basis, with the room going to the first person to book.
- The new system will be an internal system with all users having full access, except for catering modifications or cancellations.
 - Only managers can acknowledge catering changes or cancellations.
- Some other capabilities of the new system should include:
 - Access for multiple people simultaneously.
 - A record of all booking data, including cancellations.
 - Accounting and marketing reports.
 - Accessibility on in-office PC and laptops.
- Revised operating costs include:
 - Hiring a booking specialist at \$65,000 per year.
 - Hiring an assistant booking specialist at \$60,000 per year.

- Miscellaneous expenses of \$9,500.

Open Items:

- Clarify the business rules for “unconfirmed” reservations and for modifying or canceling reservations.
- Determine whether to keep track of manager “acknowledgments.”
- Update cost-benefit analysis based on revised operating costs.

Detailed Notes/Transcript:

Current System

We're going to start with the current state of the system. How are conference rooms currently booked, and what is the booking process?

- Right now it's all paper. So somebody will call us and ask for, you know, just information.
- Normally, they'll come in when they're really ready to do something because they want to see the rooms, right? Visualize it. Because it's probably a big event. It could be a wedding. It could be some sort of conference, and they just want to visualize where everything is going to be.
- So we'll take down their information, and then we'll set aside those rooms for the days (the day or days) that they've said they want them reserved for. So it's too simple. Things get lost. It's all on paper. And it's just kind of a mess at the moment.

Where do you store that information? How do you keep track of all that?

- All in a filing cabinet and sticky notes on a calendar on the wall.

So, what are the current challenges that you're facing with this process, and what would you hope a new system would improve?

- Well, we get too many double-bookings because things aren't filed properly. Making errors that are things that shouldn't be errors: as far as you know, they wanted two rooms; we only gave them one. Didn't coordinate catering very well. because it wasn't communicated very well. So I want to be able to solve all those, right?
- Make it so it's automated in some way so that we can ensure we don't double-book and get all the information we need in order to successfully reserve rooms.

Business Goals

Moving on to the business goal question. So, of these goals, which one do you think is the most important to you right now for the system? To maximize the revenue, to improve system accuracy, system efficiency, or do you think all of these are equally important to you?

- Well, they go hand in hand. The more efficient and the more effective you are, error-free, the better you're going to put on a good face, so to speak, right? And those revenues will come.

- So they're related, but you have to have good customer service and be accurate before you can ever deal with the revenue.

Do you have any other goals that are as important as these?

- You know, the biggest thing is just to try and make it as efficient and as easy to use as possible.

Future System Functionality

And then, for future system functionality, what should customers be able to do on the system? Is booking a conference room the only goal?

- Well, first of all, customers don't have access to the system. This is an internal process because they don't understand all the intricacies of how to plan an event, right? So we're there to help them through that. So being able to identify which room(s), you know, multiple rooms, if there's catering involved, and being able to track their payments and so forth is very key to this.

And then how should the system handle scheduling conflicts, or like double bookings?

- Well, hopefully, when you look for availability, you're not showing rooms that aren't available during the time this customer wants to book. So that's the whole goal, right? Only show what's available.
- Kind of like when somebody uses Outlook and tries to look for availability for a meeting. Like you guys did for your project charter, right? What time is available?

Reporting

For the data and outputs, what type of reports and performance metrics should the system consistently generate?

- We want to be able to specifically look at it from an accounting perspective, like revenue per room, revenue returning for a period of time, and which rooms. And then marketing, right? Which rooms are being reserved the most, which ones aren't just kind of looking at? You know that, that kind of usage, if you will.

Should we also keep cancellations in mind?

- Yeah. That's a good point. Yeah. Cause we're not going to delete a cancellation. We're just going to mark it as canceled so we can do reporting on it as you suggest.

Nonfunctional Requirements

What devices should this system be accessed on? Like, only PCs in the office, or should it be accessed on your phone?

- Just on PCs and laptops in the office.

Who should have access to the system? And then, should there be different clearances depending on position within the office?

- Yeah, anybody granted access will have to have an ID and a password. If they have an ID and password, they'll have full access to the system, with one exception.
- A manager has what we call "acknowledgment power." And what that means is that if there's catering involved with a reservation, and if there's a change or cancellation to that reservation, the manager must acknowledge that change or cancellation because they're the liaison with the caterer. So they need to be able to provide them with that information.
- So again, they're not approving it, but when it's submitted, it's like you go to the store and you want to make a return, and the clerk at the counter can't take it; they've got to get somebody else to approve it. We're acknowledging it, put in their ID and password to say, Yep, I acknowledge this. It's completed.

And should this system only recognize the amounts in U.S. dollars or should it be able to distinguish between foreign currencies?

- Just U.S.

If we were to expand the facilities, would the system be able to support new services, or should we design it for just the current conference room?

- Flexibility, right? You've got a little bit of flexibility you can put into it, but again, you've got a budget to deal with. So what can you get done in what we've got time and money available?
- Questions?

Additional Questions

A follow-up question to the reporting. How long should the historical data be? Like one year, three years?

- Right now, we're going to keep it forever until we decide that it's just too much. It's going to be a while before it becomes too much.

How many people can access the system simultaneously? So, should it be like one person can access the system at any certain time, or should it be multiple people?

- Multi-user, but you're not going to have a whole lot; you're only going to have two or three people in the system at the same time. But it should be able to handle that with no problem.

So, ideally, from the customer or whoever books the room to your system, how long should it take to be entered into the system?

- As soon as and quickly as possible. We want to be efficient. So, when you're searching for availability, we don't want to sit around for 10 minutes waiting for that result. We want to be able to do so within seconds.

So, like what if, let's say, two people book the same room at the exact same time, how would that work?

- The first one in gets it. The second one gets an error message, “Sorry, it was available, but now it's not.”
- That was a good question, by the way. What else?

Additional Information Provided by Sponsor

- All right, let me tell you some things that you didn't ask for that you need to know. All right? You need to know what information you need to track. Okay?
- So you need to know from a customer standpoint. I need to know their name. If they're part of an organization, I want to know their organization's name. I want a physical address, an email address, and a contact phone number. There may be some other things that we want to track for them personally. And we've got to protect it. Right?
- Venues, got to know the name of a room, the room size, capacity, and rental fee, which is usually between \$75 and \$100 per hour. But there will be steep discounts for multiple rooms and different types of events, which will be at the discretion of the booking specialist.
- Need to know about caterers. All right. So all we're looking for is their contact information. Okay. And then of course, which events they're assigned to.
- Then you got the reservations themselves. What room or rooms are assigned to that? What date and time does that reservation start? What date and time does that reservation or slash event end? What's the total price we're charging for this reservation? One or more rooms? Want to know the status of that reservation. It could be unconfirmed, confirmed, or canceled. It goes back to your earlier question about tracking canceled reservations. A room is considered unconfirmed when you initially create the reservation. It is confirmed only when they make their down payment. It's the first payment. So we're going to have to track payments.
- We're not processing payments. We're going to track payments. I've got another system in the hotel that will handle credit cards and similar transactions for the financials. But we still need to know which reservation it applies to, what the date is, and how much. So that if they call in and say, “Hey, this is my reservation. How much do I owe?” You can go figure it out. Calculate it based on how much we charge them initially, and reduce it by the number of payments they've made. Okay.
- So those are the things that we need to track. Okay. Everybody clear on that? All right.
- Also, some things are going to change over time. You've probably learned a lot about that over the last few weeks. What you know now will be different from what you know tomorrow or the day after, etc. Think back to when we talked about it in Chapter 2 during Planning, you could be 400% off on your cost and 60% off on your timing, right? Well, you're going to get more information, which will narrow it down. Like watching a hurricane on a weather map, right? They have this big wide swath of where it might hit land, but as they get more data and more time passes, it narrows down, and they can narrow it down pretty well. Well, same idea with the project, okay?
 - With that understanding, here's your first additional piece of information or change, if you will. If you look at your cost-benefit analysis, again, things change, things come more to light. We've decided to hire two new resources. Okay. We're going to hire a

booking specialist at \$65,000 per year. We're going to hire an assistant booking specialist at \$60,000 per year.

- Additionally, I want to include a \$9,500 miscellaneous expense budget item. All three of these should be line items in your operating costs. Okay. Before, you just had some kind of fillers, if you will, because you weren't sure. Well, now you're getting more information. So make sure you have those in there with whatever else comes along the way. You'll be making more additions as we move forward.

Follow-Up Question(s):

- How long can unconfirmed reservations remain in the system before the room is released?
 - At the discretion of the Booking Specialist.
- Should the system include backup/recovery?
 - Save that for Design, and that will be up to you to recommend.
- Are there any security policies or standards that the system should follow?
 - Protect data.
- Should the system track an audit log of all reservation changes (who made the change, what was changed, and when)?
 - Yes, but that will not be included until Design.
- Are there any business rules that determine when a reservation can or cannot be changed? As in, is there a certain number of days before an event where it can no longer be changed?
 - Not systematically.
- Should the system provide alerts or warnings when a booking approaches capacity limits or during high-demand periods?
 - No.